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FORGE AI — HARDWARE PROTOTYPE

Complete Bill of Materials (BOM) & Bench Assembly Technical Specifications

Author: Ali CNC Engineering
Project: ForgeAI Spindle Copilot
Target Silicon: ESP32-S3 (Dual-Core Xtensa)
Response Latency: < 2.0 ms (Edge E-Stop)

1. Executive Summary & System Overview

The **Forge AI Hardware Prototype** is an industrial edge acoustic monitoring instrument designed to detect high-frequency micro-chatter, spindle bearing wear, and cutting tool fracture on CNC milling spindles before catastrophic breakage occurs.

Audio is ingested at **48 kHz (24-bit Mono)** via an omnidirectional digital MEMS microphone over an I2S DMA bus, transformed into an 80-band acoustic vector using hardware-accelerated Radix-2 Real FFT, and evaluated against safety heuristics. A tripped anomaly triggers an optically isolated solid-state E-Stop circuit within **< 2.0 milliseconds**.

2. Official Bill of Materials (BOM) — Procurement Checklist

Hand this itemized list to component distributors (College Road / Electronic Market):

#	Component Name	Technical Specification	Qty	Function / Role
1	**ESP32-S3 DevKit Board**	**ESP32-S3-WROOM-1 / N16R8** (16MB Flash, 8MB Octal PSRAM, Dual Type-C)	1 pc	Main Edge Processor • Runs 48kHz I2S DMA + 2048-pt hardware FFT.
2	**INMP441 MEMS Microphone**	**Omnidirectional Digital I2S Module** (High SNR 61 dBA, Bottom ported, 24-bit)	1 pc	Direct acoustic ingestion from CNC spindle chassis.
3	**PC817 Optocoupler Breakout**	**PC817 1-Channel / 2-Channel Module** (Screw terminals, 5kV galvanic isolation)	1 pc	Galvanically isolated E-Stop trigger • Protects CNC controller.
4	**Prototyping Breadboard**	**MB-102 Full Size (830 Tie-Points)** (Do not purchase mini 400-point board)	1 pc	Provides sufficient clearance on both sides of wide ESP32-S3.
5	**Dupont Jumper Wires (M-F)**	Male-to-Female Ribbon Cable (20cm)	1 pack (40 pcs)	Module pin headers to breadboard interconnects.

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6	**Dupont Jumper Wires (M-M)**	Male-to-Male Ribbon Cable (20cm)	1 pack (40 pcs)	Breadboard rail jumpers and circuit interconnects.
7	**Visual Trip Indicator LEDs**	5mm High-Brightness Diffused Red LEDs	5 pcs	Visual E-Stop trip confirmation on test bench.
8	**Current-Limiting Resistors**	220Ω or 330Ω Metal Film (1/4 Watt, 5%)	5 pcs	Limits LED & optocoupler diode forward current.
9	**Isolated Test Power Supply**	9V Heavy Duty Battery + 9V Snap Connector Clip	1 set	Powers the fully isolated secondary side of the E-Stop loop.
10	**USB Data & Flashing Cable**	USB Type-A to Type-C (or Type-C to C) *Must support Data/Sync*	1 pc	Firmware flashing, serial console telemetry, and 5V power.

Estimated Local Market Total: **Rs. 3,000 – 3,400 PKR**

3. Electrical Pinout & Wiring Schematic

A. INMP441 Microphone → ESP32-S3 Wiring

> **CRITICAL VOLTAGE NOTICE:** Connect INMP441 VDD strictly to the **3.3V pin** of the ESP32-S3. Connecting VDD to 5V will permanently destroy the MEMS sensor.

INMP441 Pin	ESP32-S3 Pin	Signal Type	Firmware Role
VDD	**3.3V**	Power Supply	3.3V Regulated Rail
GND	**GND**	Ground Reference	Common System Ground
SD (Serial Data)	**GPIO 4**	Digital Input	I2S Direct Memory Access (DMA) Ingestion
SCK (Bit Clock)	**GPIO 2**	Digital Clock	I2S Serial Clock (BCLK)
WS (Word Select)	**GPIO 3**	Digital Sync	I2S Frame / Left-Right Clock (LRCK)
L/R	**GND**	Channel Select	Tied to GND for Left Channel (Mono Mode)

B. PC817 Optocoupler & Isolated E-Stop Trip Circuit

The optocoupler ensures physical galvanic isolation between the microcontroller and the CNC emergency stop circuit.

Optocoupler Pin	Connection Point	Description / Circuit Function
Pin 1 (Anode)	**GPIO 5 (via 330Ω resistor)**	MCU Logic High triggers the internal infrared transmitter diode.
Pin 2 (Cathode)	**ESP32 GND**	Completes primary low-voltage circuit.
Pin 4 (Collector)	**+9V (Red Lead of 9V Battery)**	High-side isolated voltage supply.
Pin 3 (Emitter)	**LED Anode (+) → 330Ω → 9V GND (-)**	When tripped, current conducts from 9V battery through the Red LED to ground.

> **CIRCUIT BEHAVIOR VERIFICATION:** Under normal spindle operation, GPIO 5 remains LOW (0V). When chatter harmonics exceed safety envelopes, the ESP-DSP firmware pulls GPIO 5 HIGH (3.3V). The phototransistor saturates within < 2.0ms, illuminating the Red LED and opening/closing the CNC E-Stop contact.

4. Step-by-Step Bench Assembly Instructions

1. **Mount ESP32-S3:** Insert the ESP32-S3 dev board across the center divider of the MB-102 breadboard so row pins on both sides remain accessible.
2. **Wire INMP441:** Connect GND and 3.3V first. Connect SCK → GPIO 2, WS → GPIO 3, and SD → GPIO 4. Ensure L/R pin is firmly strapped to GND.
3. **Wire Optocoupler Primary:** Connect ESP32 GPIO 5 in series with a 330Ω resistor to PC817 Pin 1 (Anode). Connect Pin 2 (Cathode) to ESP32 GND.
4. **Wire Optocoupler Secondary:** Connect the 9V battery red lead (+) to PC817 Pin 4 (Collector). Connect PC817 Pin 3 (Emitter) to the Red LED long leg (Anode). Connect the short leg (Cathode) through a 330Ω resistor back to the black lead (-) of the 9V battery.
5. **Firmware Flash:** Connect USB-C data cable from PC to the ESP32-S3 **USB / COM** port. Compile and flash `Edge Firmware/esp_dsp_firmware.cpp` using ESP-IDF v5.x.
